

Dynamic Ensemble of Deepfake Detectors Conditioned on CLIP Features

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Abstract

Deepfake image detection framework that combines the outputs of multiple pre-trained detectors using dynamically predicted importance weights is proposed. The final prediction is computed as a convex combination of detector scores, with weights derived from CLIP image embeddings, enabling the ensemble to adapt to different types of synthetic content. To support evaluation, we construct a dataset of approximately 50k images, including four real-image sources and fake images generated by ten modern synthesis tools spanning GANs, diffusion models, and commercial systems. We set up fourteen publicly available detectors and benchmark them individually and within our ensemble. Extensive experiments show that the proposed model outperforms individual detectors and simpler baselines, particularly under image degradations and when generalizing to unseen generators. Ablation studies further confirm the benefits of dynamic weighting over static combinations or CLIP-only predictors.

1. Introduction

The recent developments in generative models have undermined the credibility of visual media. Images once assumed to be trustworthy can now be effortlessly manipulated using modern synthesis techniques. Contemporary image-generation models, such as generative adversarial networks and diffusion-based architectures, can produce photorealistic face images or manipulate real ones with such precision that the results are often indistinguishable from authentic images, even for trained experts.

While this progress has opened new opportunities in creative expression, education, and personalised content creation, it also introduces considerable societal and security risks. Face-image generation can be weaponised to fuel misinformation, interfere with democratic processes, or facilitate identity deception in financial systems. Moreover, the widespread availability of these tools enables even unskilled users to produce realistic synthetic media at scale. Text-to-image generation, in particular, allows adversaries to construct highly specific imagery aligned with their intended narrative.

Consequently, there is a high demand for tools that

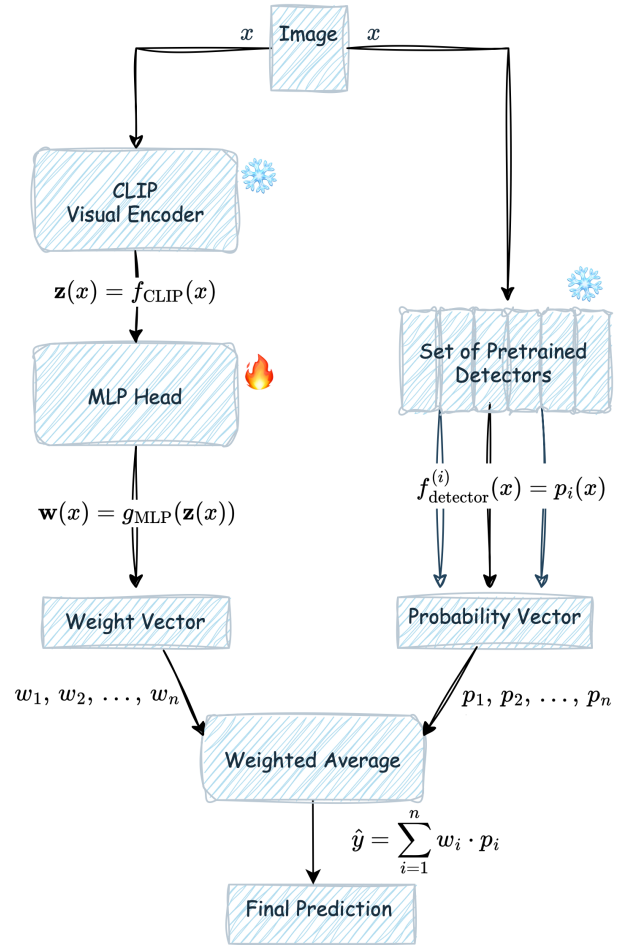


Figure 1. **The proposed architecture of the ensemble model.** For a given image, a bank of pre-trained deepfake detectors is executed, producing a probability vector. The final “fake/real” prediction is computed as a weighted average of the individual detector probability scores. The model is trained to estimate the importance weights dynamically from image embeddings produced by the CLIP visual encoder. Models following this architecture are denoted Ens-Small and Ens-Final.

help governments, fact-checkers, or journalists establish authenticity even when images have been deliberately altered to evade detection. This work aims to support experts in answering a question that is becoming increas-

ingly common in social media contexts or journalism: Is this real or not?

The research landscape of deepfake image detection is highly active, with a continual emergence of new detector models. However, their accuracy varies substantially. A key challenge is poor generalization: many detectors fail when confronted with fake content that differs from the data seen during training. Our approach is to aggregate a diverse set of available detectors and construct an ensemble that leverages all of them simultaneously. Since many detectors appear to specialize in particular families of generators (*e.g.*, GANs or diffusion models), we propose the ensemble architecture shown in Fig. 1, where the final probability score is computed as a convex combination of individual detector outputs. The corresponding weights are predicted dynamically from image embeddings produced by a foundation model, namely CLIP [22].

Our contributions are as follows:

- We constructed a diverse evaluation dataset made up of about 50k images, containing 4 different real datasets and realistic face images manipulated or fully generated by 10 modern synthesis tools, including generative adversarial networks, diffusion models, and commercial image generators.
- We set up 14 deepfake image detection models sourced from publicly available GitHub repositories.
- We proposed and developed an ensemble detection model that combines outputs of a set of pre-trained detectors with image features.
- We performed an ablation study comparing the proposed ensemble architecture with models that either rely solely on detector outputs combined using static weights or use only image embeddings without detector outputs.
- We conducted extensive experiments evaluating the detectors’ robustness to common image transformations (degradations that occur in social media) and their ability to generalize to previously unseen generators.

The remainder of the paper is structured as follows. The related work is summarized in Sec. 2, the proposed method is presented in Sec. 3, the experiments are given in Sec. 4, and finally Sec. 5 concludes the paper.

2. Related Work

2.1. Deepfake production techniques

Advanced face manipulation began with techniques such as Face2Face, which used 3D models to transfer expressions from one person to another in real time [29]. The field then shifted significantly with the rise of deep learning and deepfakes, where autoencoders and GAN-based models enabled face swapping with much higher realism [17]. For example, SimSwap improved face swapping by injecting source identity into the target’s features while preserving target attributes such as expression and gaze direction [4]. More recently, diffusion models have been introduced into face manipulation tasks, achieving

improved performance compared to GAN-based methods [14].

2.2. Deepfake detectors

The detection of fake images has become a critical research area in recent years due to rapid advances in generative technologies. These methods can modify existing images or synthesise them entirely. As generative models continue to improve in visual quality, fake images have become increasingly difficult for regular, untrained observers to identify [10]. Notably, even experts often struggle, sometimes misclassifying authentic images as generated and attributing imperfections made by human artists to AI [10].

Classifiers trained in-distribution (*i.e.*, on data from a specific image-generator class) achieve high accuracy but often fail when encountering images from unseen generative models [20]. Since new generative models emerge frequently, there is a growing need for detectors capable of identifying fake images produced by previously unseen methods. This property is referred to as *generalization*, and it reflects a detector’s ability to perform well across a range of generation techniques and manipulation types. Another important property is *robustness*, which denotes a model’s ability to maintain accuracy under transformations such as resizing, compression, or adversarial noise. To improve generalization and robustness, researchers have introduced heavy augmentations, including JPEG compression and blurring [30].

Wang et al. [30] showed that standard CNN classifiers trained on images generated by models such as ProGAN can generalise effectively across different GAN-based images, likely due to distinctive and persistent low-level artifacts shared across many GAN generations. However, detectors trained on GAN-generated images often fail when applied to diffusion-generated images [23]. These findings laid the groundwork for exploring more generalisable detection methods.

There are approaches that do not rely on training on large datasets of fake images in order to avoid overfitting. Shiohara et al. [27] focus on detecting blending artifacts that are common in face-swap deepfakes, but absent in fully synthetic images. Another method, DIRE [32], reports strong generalization based on the observation that synthetic images are easily invertible by diffusion models, whereas real images are not.

In recent research, leveraging a large foundation model as an image encoder, such as CLIP, has demonstrated notable generalisation across multiple deepfake-production techniques and strong robustness, highlighting the advantages of large-scale pretraining [7, 34].

2.3. Ensemble models

Ensemble models are a well-established strategy for improving classification accuracy by combining the outputs of multiple classifiers and reducing the overall bias of the final prediction.

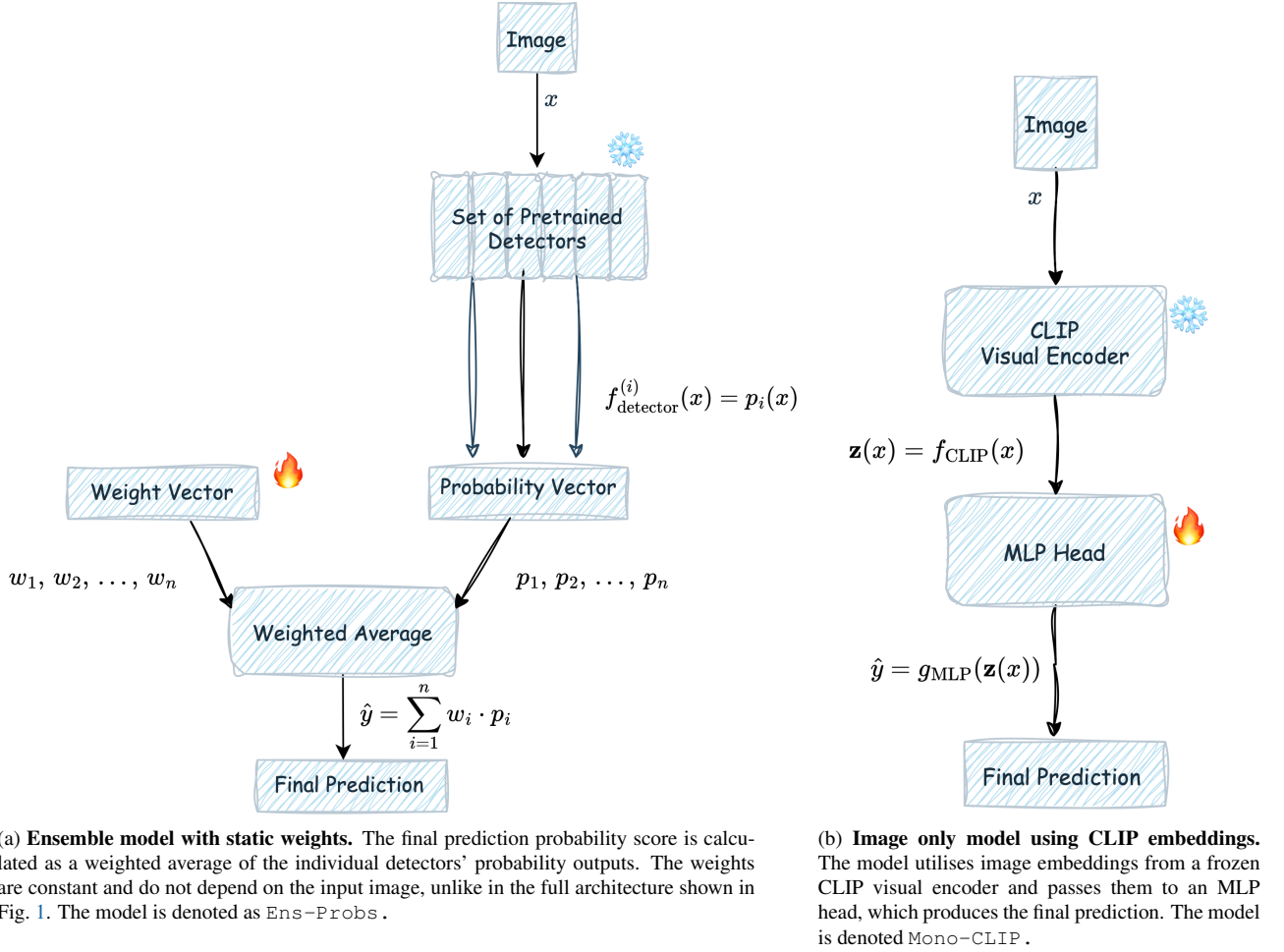


Figure 2. Diagrams of two baseline detection models explored in this work.

An early study regarding deepfake detection by Panigrahy et al. [19] employed an ensemble of 13 CNN architectures including Xception, DenseNet121, and ResNet variants with a majority voting mechanism for identifying manipulated images created via splicing and copy-move operations. This approach demonstrated significant improvements over single-model solutions.

More recently, computationally efficient ensemble methods suitable for mobile deployment have been proposed, combining lightweight CNN architectures like EfficientNetB0 and MobileNetV2, showing promising generalisation capabilities across various GAN-based generators [37]. However, these results primarily focused on GAN-generated images, known for their distinctive and detectable artifacts.

Another ensemble-based approach, MaskCLIP [31], integrates pretrained models such as CLIP and MAE (Masked Autoencoders) to detect high-level semantic inconsistencies and low-level reconstruction anomalies simultaneously. Furthermore, MaskCLIP employs learnable continuous prompts and attention mechanisms to combine diverse feature sources.

3. Method

The primary motivation for the ensemble model is to enhance the robustness and generalization of existing fake-image detectors while leveraging their learned knowledge. Individual detectors may exhibit limited generalization due to their training data or architectural constraints. Nevertheless, their outputs contain valuable information, and our goal is to exploit this by combining the predictions of individual detectors into a collective decision mechanism.

To achieve this, we introduce an additional degree of freedom by incorporating semantic image understanding through the CLIP visual encoder [22] to condition the contribution of each detector. The learned weighting mechanism adapts dynamically, assigning more or less trust to individual detectors depending on the input image. Conditioning the weighting system on CLIP features enables the model to recognize when a detector is unreliable or confident with respect to specific artifacts and to accordingly decrease or increase that detector's influence in the final prediction. This approach allows us to leverage transfer learning in two complementary ways: by exploiting CLIP's visual understanding and by utilizing the outputs

of existing fake-image detectors. Moreover, the architecture provides interpretability by producing per-image detector weights.

3.1. Proposed architecture

The architecture of the proposed model is shown in Figure 1. It is designed to weight the outputs of pre-trained detectors according to visual cues extracted from the image by the CLIP visual encoder. For a given input image, all pre-trained detectors are executed, producing scores that form a probability vector. In parallel, semantic features using a frozen pre-trained CLIP visual encoder are extracted. These features are further processed by a small multi-layer perceptron head, which produces the weight vector. The final prediction score is computed as the dot product between the weight and probability vectors. The output is therefore a convex combination of the individual detector outputs, as the weights are non-negative and sum to one.

Model Formulation Let $x \in \mathbb{R}^{H \times W \times 3}$ be an input image. The CLIP visual encoder extracts a feature representation $f_{\text{CLIP}}(x) \in \mathbb{R}^d$, which is passed to an MLP head g_{MLP} that maps it to a weight vector

$$\mathbf{v}(x) = [w_1(x), w_2(x), \dots, w_n(x)] = g_{\text{MLP}}(f_{\text{CLIP}}(x)),$$

which is normalised with the softmax function

$$\mathbf{w}(x) = \text{softmax}(\mathbf{v}(x)),$$

then $w_i(x) \geq 0, \sum_{i=1}^n w_i(x) = 1$.

Simultaneously, the image is passed to a fixed set of n pretrained detectors, each defined as a function $f_{\text{detector}}^{(i)}: x \rightarrow \langle 0, 1 \rangle$. Each detector outputs a probability score (indicating the likelihood of belonging to the “fake” class) for the input image:

$$p_i(x) = f_{\text{detector}}^{(i)}(x), \quad \text{for } i = 1, \dots, n.$$

These individual outputs form a probability vector:

$$\mathbf{p}(x) = [p_1(x), p_2(x), \dots, p_n(x)] \in [0, 1]^n.$$

The final prediction is computed as the dot product between the weight vector and the vector of detector probabilities, producing the final probability score

$$\hat{y}(x) = \mathbf{w}(x)^\top \mathbf{p}(x) = \sum_{i=1}^n w_i(x) \cdot p_i(x) \in [0, 1],$$

where $\hat{y} \approx 1.0$ indicates full confidence in the fake class and $\hat{y} \approx 0.0$ indicates full confidence in the real class.

3.2. Baseline architectures

The proposed model is ablated in two ways, resulting in the following baseline variants.

Static Ensemble Detector. We implemented an ensemble with static weights shown in Figure 2a using only detector outputs (without image features) to assess whether the image-based features are necessary for optimal weighting.

Let $\mathbf{w} = [w_1, w_2, \dots, w_n] \in \mathbb{R}^n$ be a fixed weight vector, where the weights are constrained that $w_i \geq 0$ and $\sum_{i=1}^n w_i = 1$. The rest is the same as the full model, so

$$\hat{y}(x) = \mathbf{w}^\top \mathbf{p}(x) = \sum_{i=1}^n w_i \cdot p_i(x) \in [0, 1].$$

Note that the weight vector is constant, *i.e.* does not depend on the input image.

To learn the constant weight vector, we employed an evolution-based optimisation¹, which turned to be more accurate than gradient descent methods. The static ensemble was faster to train, as we were able to produce a decent detector in less time than it takes to train one epoch of the full model.

Image Detector. Secondly, we implemented a monomodal detector from Fig. 2b that relies exclusively on image information. Specifically, a classification head is applied to image embeddings extracted by the CLIP visual encoder. No pretrained detectors are involved in the model.

Formally, given an input image $x \in \mathbb{R}^{H \times W \times 3}$, the CLIP visual encoder extracts a feature vector $f_{\text{CLIP}}(x) \in \mathbb{R}^d$. This feature is passed through a classification head h_{MLP} , producing a scalar prediction:

$$\hat{y}(x) = h_{\text{MLP}}(f_{\text{CLIP}}(x)) \in [0, 1].$$

4. Experiments

4.1. Datasets

For our experiments, we collected a dataset of human face images. The dataset contains both manipulated images (*e.g.*, face swap, face reenactment) and fully synthetic face images produced by various generators. To generate the images, we provided a detailed prompt for the sake of diversity, *e.g.* “close-up on face of a woman with gray hair, alabaster skin, and amber eyes, having zen expression set against a countryside background, with studio lighting.” Real images were sampled from several datasets.

All images were subjected to the same preprocessing pipeline: face detection, cropping, and resizing to 224×224 px.

Training and validation set. The composition of our training set is given in Tab. 1. We have two variants; the Large set and the Small set. Intermediate and baseline models were trained on the small set, the final model was trained on the Large set. Fake images were mostly

¹We used the `differential_evolution` implementation from `scipy.optimize`.

Table 1. **Training dataset composition.** The table shows the training dataset for the final ensemble model and smaller experimental variants. The validation set corresponds to a proportion of the original data.

Source / Config.	Large	Small	Label
FF++	13,557	2,500	Fake
RV	7,690	2,500	Fake
Flux	50	–	Fake
DF-IF	50	–	Fake
WFIR	10	–	Fake
LAION	18,878	5,000	Real
Total	40,235	10,000	–
Validation set	5%	10 %	–
Augmentations	45%	45 %	–

taken from Face Forensics++ [24] and generated by Realistic Vision (RV) model [5]. Real images for training were sampled from LAION dataset [35].

To prevent overfitting of the model and improve robustness to image degradations, we augmented 45% of the images in each epoch using techniques such as random quality level up to 60% JPEG compression, rotation, grayscale conversion, sharpening, Gaussian blur, affine transformations, color jittering, horizontal and vertical flips, and random cropping. The validation set is non-overlapping 5% and 10% for Large and Small sets, respectively. The rate of augmentation is the same on the validation set.

Test dataset. The composition of the test set is given in Tab. 2. It is on purpose different from the training set in both fake and real parts to test the model generalization to unseen content. For each generator and the real subset, we collected 1000 images. The images were either generated using public models (*e.g.*, via Hugging Face) or sourced from benchmark datasets such as DeepFakeBench [24] and DF40 [33]. Real images were sampled from publicly available datasets, including IMDB-WIKI [25], CelebA [15], FFHQ [13]. Note that the real images are taken from a different dataset than the training set, which is a challenging situation, as the detector might overfit to possible biases in the real data.

4.2. Detectors

The pre-trained deepfake detectors used in our ensemble models are listed in Tab. 3. Multiple publicly available detectors of various approaches that span years 2020–2026 were used.

We tested several of our models: *Ens-Small* – the model of architecture in Fig. 1, and two ablated baselines *Ens-Props* – of architecture in Fig. 2a, and *MonoCLIP* – see Fig. 2b. They were trained on the small training set detailed in Tab. 1, and using transformer ViT-B/32 of the OpenCLIP image encoder [11]. The final ensemble model *Ens-Final* was trained on the large training set and with a larger tranformer backbone ViT-L/14.

Table 2. **Test dataset.** The table lists all generators and real image sources used for evaluation, along with their respective origins and the reference names used throughout this work.

Generator	Source	Ref. Name
StyleGAN3	DF40 dataset [33]	StyleG3
FaceForensics++	FaceForensics++ [24]	FF++
PixArt-alpha	Generated [3]	PixArt
Realistic Vision v6	Generated [5]	RV
Stable Diffusion XL	Generated [21]	SD-XL
Kandinsky 3.0	Generated [1]	Kndsky
FLUX-dev	Generated [2]	Flux
DeepFloyd IF	Generated [8]	DF-IF
WhichFaceIsReal	DF40 dataset [33]	WFIR
MidJourney v6	DF40 dataset [33]	MidJ
Real images	IMDB + Celeba + FFHQ	Real

Table 3. **Pre-trained detectors.** The table lists all pre-trained detectors that were used in our ensemble models.

Detector Name	Year	Publication
CNND, CNND2 [30]	2020	CVPR
FreqD [9]	2020	ICML
GramNet [16]	2020	CVPR
Fusing [12]	2022	ICIP
DIMD-latent, DIMD-gan [6]	2023	ICASSP
UnivFakeD [18]	2023	CVPR
RPTC [36]	2023	ArXiv
DeFake [26]	2023	SIGSAC
NPR [28]	2024	CVPR
ClipBased, Corvi2023 [7]	2024	CVPRw
DeepfakeD [34]	2026	WACV

All models were trained to minimize binary cross-entropy loss. The final model was extensively trained for approximately five days using NVIDIA A100-SXM4 GPUs with 40GB of memory. Training epochs varied in duration, typically requiring between 45 minutes to two hours depending on computational resource utilisation and the complexity of the training scenarios.

4.3. Results

We evaluated all pretrained detectors and our models on subsets of our test set. Testing was performed under two scenarios: original images and degraded images produced by random cropping (up to 5/8), resizing (down to 0.5), and JPEG compression (down to 60%). The latter setting simulates typical downstream processing on social-media platforms. The results are shown in Tab. 4 and Tab. 5, respectively.

The tables report classification accuracy. All test subsets are balanced, containing 1000 real and 1000 fake images, so the chance level is 0.5. We gray out entries where the generators in the training and test sets overlap. We refer to these as “in-distribution” cases, and their accuracy values are excluded from the row-average calculations. While other detectors may encounter in-distribution data during evaluation and benefit from this overlap, our

Table 4. **Accuracy of detectors across generative models (original images).** Accuracy is evaluated per generator using balanced sets of fake and real images. The table shows detectors (rows) evaluated against various generative models (columns). Greyed-out numbers correspond to generators included in their training set; these values are excluded from the row-average calculations.

Detector	DF-IF	MidJ	RV	SD-XL	Flux	Kndsky	PixArt	FF++	StyleG3	WFIR	Row Avg.
CNND	0.53	0.76	0.47	0.47	0.52	0.47	0.56	0.47	0.59	0.91	0.57
CNND2	0.50	0.54	0.50	0.50	0.50	0.50	0.51	0.50	0.50	0.56	0.51
Corvi2023	0.45	0.94	0.95	0.92	0.56	0.86	0.95	0.45	0.62	0.45	0.72
DIMD-gan	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.96	1.00	0.60
DIMD-latent	0.45	0.92	0.95	0.91	0.58	0.81	0.95	0.45	0.45	0.45	0.69
DeFake	0.58	0.55	0.70	0.72	0.63	0.67	0.70	0.52	0.49	0.70	0.63
FreqD	0.46	0.32	0.64	0.62	0.40	0.50	0.58	0.50	0.72	0.78	0.55
Fusing	0.84	0.83	0.48	0.47	0.54	0.48	0.61	0.47	0.88	0.73	0.63
GramNet	0.78	0.73	0.57	0.29	0.74	0.78	0.54	0.35	0.79	0.79	0.64
NPR	0.65	0.88	0.53	0.51	0.86	0.82	0.78	0.41	0.88	0.53	0.69
RPTC	0.61	0.57	0.70	0.57	0.60	0.78	0.72	0.33	0.80	0.80	0.65
UnivFakeD	0.49	0.49	0.49	0.49	0.49	0.52	0.49	0.65	0.81	0.57	0.55
ClipBased	0.66	0.40	0.87	0.57	0.70	0.87	0.58	0.86	0.88	0.57	0.70
DeepfakeD	0.52	0.58	0.50	0.54	0.51	0.62	0.65	0.93	0.98	0.51	0.63
Ens-Small	0.64	0.62	0.96	0.63	0.70	0.95	0.84	0.89	0.96	0.79	0.77
Ens-Probs	0.76	0.88	0.93	0.61	0.82	0.93	0.89	0.47	0.93	0.92	0.84
MonoCLIP	0.69	0.87	0.88	0.83	0.72	0.80	0.84	0.85	0.73	0.68	0.76
Ens-Final	0.76	0.71	0.92	0.91	0.83	0.91	0.91	0.93	0.94	0.91	0.88

Table 5. **Accuracy of detectors on transformed images simulating social media degradation.** Images undergo random cropping, resizing, and JPEG compression to assess robustness under realistic post-processing conditions. The table shows detectors (rows) evaluated against various generative models (columns). Greyed-out numbers correspond to generators included in their training set; these values are excluded from the row-average calculations.

Detector	DF-IF	MidJ	RV	SD-XL	Flux	Kndsky	PixArt	FF++	StyleG3	WFIR	Row Avg.
CNND	0.51	0.52	0.50	0.50	0.50	0.50	0.51	0.50	0.52	0.51	0.51
CNND2	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
Corvi2023	0.57	0.67	0.61	0.64	0.58	0.70	0.56	0.59	0.80	0.66	0.64
DIMD-gan	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.51	0.50	0.50
DIMD-latent	0.51	0.63	0.64	0.59	0.53	0.62	0.53	0.51	0.58	0.58	0.57
DeFake	0.59	0.63	0.65	0.67	0.48	0.73	0.67	0.52	0.41	0.61	0.60
FreqD	0.50	0.61	0.46	0.55	0.49	0.59	0.62	0.46	0.46	0.54	0.53
Fusing	0.54	0.54	0.50	0.52	0.53	0.50	0.53	0.50	0.62	0.52	0.53
GramNet	0.53	0.51	0.52	0.52	0.51	0.52	0.52	0.47	0.47	0.51	0.51
NPR	0.50	0.50	0.50	0.50	0.50	0.51	0.50	0.50	0.50	0.50	0.50
RPTC	0.60	0.58	0.42	0.58	0.61	0.47	0.49	0.55	0.54	0.58	0.54
UnivFakeD	0.48	0.48	0.48	0.48	0.48	0.52	0.49	0.61	0.62	0.49	0.51
ClipBased	0.58	0.54	0.58	0.58	0.56	0.59	0.58	0.59	0.59	0.54	0.57
DeepfakeD	0.54	0.59	0.47	0.55	0.52	0.64	0.64	0.89	0.93	0.47	0.62
Ens-Small	0.61	0.64	0.75	0.68	0.58	0.79	0.66	0.86	0.88	0.55	0.67
Ens-Probs	0.57	0.57	0.60	0.57	0.55	0.59	0.56	0.50	0.51	0.56	0.56
MonoCLIP	0.54	0.55	0.81	0.57	0.55	0.58	0.55	0.79	0.73	0.51	0.57
Ens-Final	0.68	0.66	0.89	0.76	0.69	0.87	0.85	0.98	0.95	0.54	0.82

models are strictly tested on unseen images, ensuring a fair assessment of generalisation.

The proposed **Ens-Final** outperforms all competing methods by a large margin on average accuracy. The margin is grater for the the degraded images. The model is performing the best for many individual subsets.

For the smaller ensemble models, it is seen on Tab. 4

that our baseline ensemble model **Ens-Probs** exhibits the best average accuracy, while it does not achieve the top accuracy for any single generator. The result highlights that even without image features, the ensemble technique allows model to generalise effectively across unseen generators. However, when subjected to the degraded images, its accuracy drops slightly above random chance, reveal-

ing a significant lack of robustness against post-processing transformations, see Tab. 5.

On the other hand, as indicated in Table 5, the `Ens-Small` model maintains strong performance under image degradations, achieving the highest row-average accuracy among the models trained on the smaller datasets (`Ens-Probs`, `MonoCLIP`). The image only model `MonoCLIP` has a comparable average accuracy with `Ens-Small` on the original, but much lower on the degraded images. This result demonstrates the effectiveness of combining multiple pretrained detectors when models access both image features and detector outputs.

Interestingly, the architectures of our `MonoCLIP` and the `ClipBased` detector [7] are very similar, but there are still differences in accuracy. Since both models leverage CLIP embedding, we attribute the difference to a different training datasets.

Among other detectors, `DeepfakeD` and `DeFake` also demonstrate stable performance across both original and transformed images. Notably, `DeepfakeD` average accuracy remains nearly unchanged (0.63 to 0.62) under augmentations, and it remains the best-performing detector for `StyleG3` and `FF++`. In contrast, several models `CNND2`, `CNND`, `UnivFakeD`, and `FreqD` operate close to random guessing levels and consistently achieve row averages between 0.50 and 0.55. Other models `DIMD-gan`, `DIMD-latent`, `Fusing`, `GramNet`, `NPR`, and `RPTC` experience significant performance degradation under transformations, approaching random chance levels.

4.4. Contribution of individual detectors

To understand how the final ensemble model `Ens-Final` allocates decision responsibility across detectors, we analysed the per-image weights assigned by the MLP head conditioned on CLIP embeddings. Since the model computes a different set of weights for each image, we evaluated 100 samples per generator and calculated the mean and variance of the assigned weights. Additionally, detectors with weights lower than 0.01 were omitted from the figures for clarity, as their contribution to the final decision is negligible.

The dynamic weights predicted by `Ens-Final` model are shown in Fig. 3, again for two scenarios – original and images distorted by common transformations. The figure shows the weights of the static ensemble `Ens-Prob` for comparison.

It is seen that the dynamic weights differ from the static weights, so the model indeed learned to use the image features. Moreover, the dynamic weights differ for generators and real images for both original and degraded images. On the other hand, we can observe certain similarities between similar models, e.g. `Realistic Vision` and `SD XL`. Interestingly, detector `DeFake` seemed to be used for many generators. For `FF++` images, the most important was `CLIPBased`, which was the most useful for `StyleGAN3` and used more or less in all tested subsets. To

recognize real images, the most important detector was `DIMD-gan`.

Note that the weights are sparse. Weights of many detectors are always zero. Those detectors can be dropped to speed the evaluation up.

5. Conclusions

In this work, we conducted a comprehensive evaluation of fourteen publicly available deepfake detectors on a curated dataset of facial images comprising both real samples from multiple sources and synthetic samples generated by modern generative models. Building on these findings, we proposed an ensemble architecture that integrates the outputs of pretrained detectors and assigns them adaptive importance weights predicted directly from image features. Our experiments demonstrate that this approach consistently outperforms both individual detectors and simpler ensemble baselines, particularly in challenging scenarios where input images undergo degradations such as compression.

An analysis of the learned dynamic weights further revealed that, for certain generator families, only a subset of detectors contributes meaningfully to the final decision, while others receive negligible or effectively zero weight. Although this insight highlights a potential path toward computational efficiency, the current method requires running all detectors during inference, resulting in latency on the order of tens of seconds, since all the detectors do not fit into a GPU memory. Parallel computation on multiple GPUs will reduce latency. Future work will therefore focus on leveraging the discovered weight patterns to prune consistently uninformative detectors and reduce computational cost. Moreover, the observed generator-dependent weighting behavior opens promising opportunities for model attribution, which we plan to explore in subsequent research.

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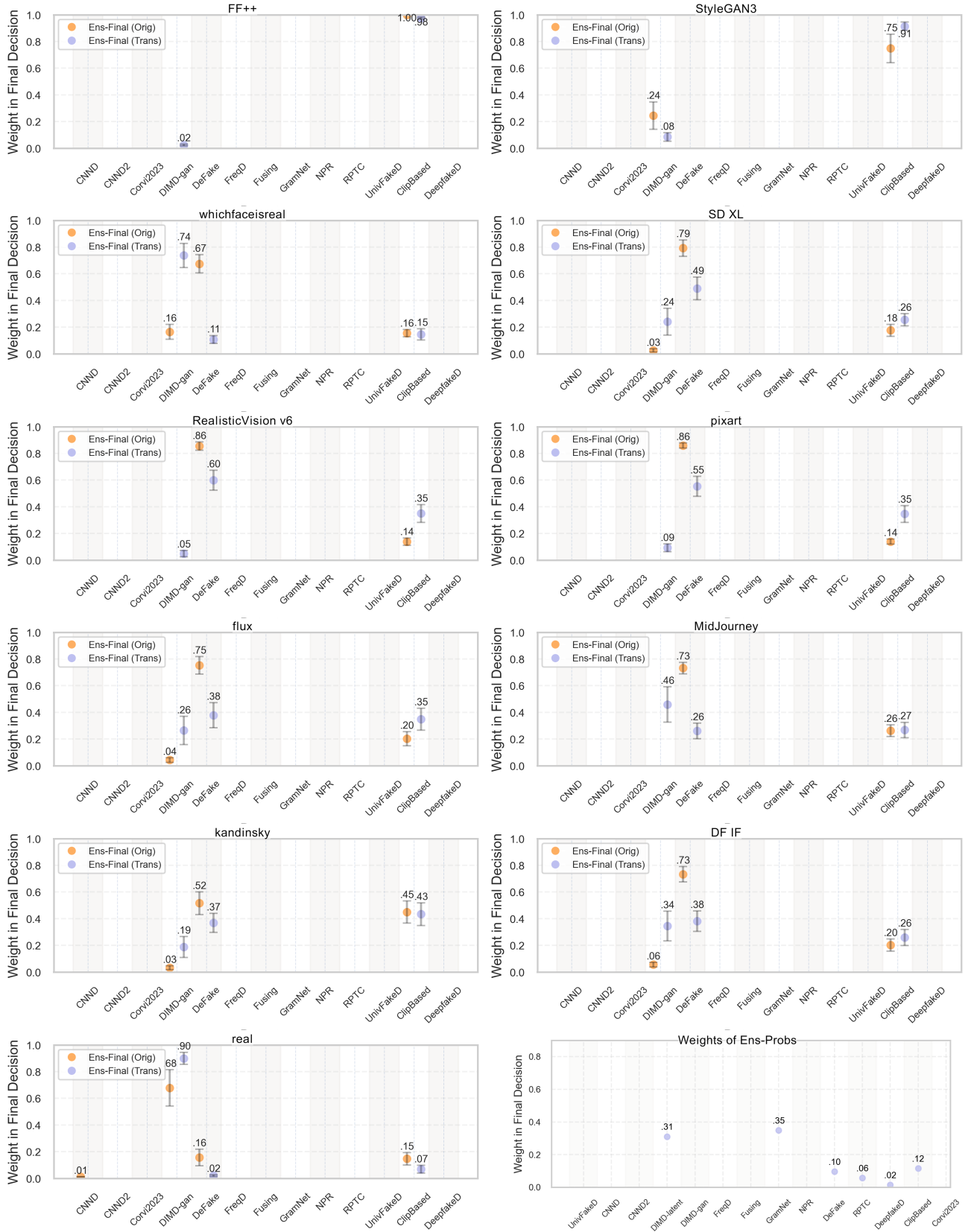


Figure 3. Importance weights of the full Ens-Final model calculated for both original (Orig) and degraded/transformed (Trans) images across individual generators and real test subsets. Means and standard deviations are computed over 100 random images. Static weights of Ens-Probs are shown for comparison.

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